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A robust periodic capacitated arc routing problem for urban waste collection considering drivers and crew's working time

Erfan Babaee Tirkolaee [a,*], Iraj Mahdavi [a], Mir Mehdi Seyyed Esfahani [b]

[a] Department of Industrial Engineering, Mazandaran University of Science & Technology, Babol, Iran
[b] Department of Industrial Engineering, Amirkabir University of Technology, Tehran, Iran

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ABSTRACT

In this paper, a novel mathematical model is developed for robust periodic capacitated arc routing problem (PCARP) considering multiple trips and drivers and crew's working time to study the uncertain nature of demand parameter. The objective function of the proposed model aims to minimize total traversed distance and total usage cost of vehicles over a planning period. To solve the problem, an improved hybrid simulated annealing algorithm (SA) is developed based on a heuristic algorithm and an efficient cooling equation. It has been proved that the performance of the proposed algorithm is acceptable in comparison with the exact solution method. Finally, the results have shown the effects of different uncertainty level of the demand parameter on the problem to be considered as a managerial overview in decision making process under uncertainty.

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1. Introduction

Nowadays, generating different types of solid wastes and manifestation of their associated social, economic and environmental inconsistencies create numerous complications for urban service management in regard to the collection, transportation, processing and disposal of such wastes. An optimal planning in collecting, management, and disposal of urban waste are one of the most important strategies that can yield multiple improvements in both total cost and health.

Due to the fact that 60–80 percent of costs of solid wastes management are related to collecting and transporting them (Lifset, 1992), the evaluation of underlying collection and transportation system plays a significant role in reducing and solving the problems of urban service management (Golden et al., 2002). The wastes should be collected, transported and disposed in the least time, through the best method, and directly from household areas to disposal sites. Based on the above discussions, the significance of an optimal system of waste collection becomes more highlighted. Therefore, selection of optimal policy of waste collection has a significant impact on reducing costs.

In routing problems related to the urban wastes collection, two categories of problems are defined (Golden et al., 2002). In the first category, there is a predefined node series with positive demand and the objective is to find the best tours which cover all nodes-Capacitated Vehicle Routing Problem (CVRP). In the second category, a series of predefined edges is existed with positive or zero demand and the objective is to find the best tours covering all required edges (edges with positive demand)-Capacitated Arc Routing Problem (CARP).

Although VRP has been studied widely in the literature with numerous applications (Tirkolaee et al., 2017; Mirmohammadi et al., 2017; Alinaghian et al., 2014), the CARP has more appropriate application in studying waste collection and its routing problems (Babaee Tirkolaee et al., 2016). So, the second category is the problem being studied in this research. In this problem, wastes are along the edges (i.e. streets or alleys with demands). In addition, the capacity of vehicles is limited and when vehicles traverse the required edges, their remained capacities decrease. Drivers and crew's working time is another important parameter which has a great impact on determining the number of required vehicles besides the capacity constraints. The objective is defined to determine the optimal routes of serving the required edges based on a daily planning in a week.

In the following, important research in the literature of CARP for urban waste collection has been reviewed and described.

Chu et al. (2005) introduced a periodic capacitated arc routing problem (PCARP) in order to provide a weekly planning horizon. They proposed a mathematical model based on a mixed integer linear programming (MILP) and solved the problem through two innovative methods. The objective was to assign a set of service

* Corresponding author.
E-mail address: e.babaee@in.iut.ac.ir (E.B. Tirkolaee).

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days to each edge in a defined network and solving the resulting routing problem for each period to minimize the size of the required fleet of vehicles and total traversing cost during the predefined time horizon.

Lacomme et al. (2005) developed a periodic capacitated arc routing problem (PCARP) for practical applications such as collecting waste. They describe few models of PCARP with a simple categorization scheme. For instance, they defined that the demand of each arc might be dependent upon the planning period or the last date of service. They used an evolutionary algorithm (EA) based on a complex intersection operator to simultaneous modify and apply operational and tactical decision makings such as service days and day trips. Pia and Filippi (2006) introduced a version of capacitated arch routing for two types of vehicles in which the first type of vehicle could unload at the base while the second type unloads its wastes into the vehicles of the first type. In their developed problem, determination of optimal routes is accompanied by optimal decision-making for two types of vehicles. They solved this problem by developing a variable neighborhood descent algorithm (VND).

Fleury and Lacomme (2004, 2005) formulated a stochastic arc routing problem (SCARP) appropriate for urban waste collection in which some parameters such as demand, vehicles travel time are considered to be stochastic. They developed some EAs to solve the SCARP. After them, Laporte et al. (2010) presented a CARP problem considering stochastic demands. They solved the problem by a neighborhood search heuristic algorithm. They could prove the efficiency of their proposed algorithm.

Ramos et al. (2013) proposed a MILP formulation for a Multi-Depot Vehicle Routing Problem with Mixed Closed and Open Inter-Depot Routes where capacity and duration constraints are taken into account. They applied their model to a real case study and could create some savings in route planning. Son and Louati (2016) proposed a generalized vehicle routing model including multiple transfer stations, gather sites and inhomogeneous vehicles in time windows for Municipal Solid Waste (MSW) collection. They could show that their model reduces both total traveling distances and operational hours of vehicles in comparison with those of practical scenarios after applying a case study to the model.

Babaee Tirkolaee et al. (2016) developed a robust CARP for a waste collection application using a hybrid simulated annealing algorithm. They showed that the performance of the proposed hybrid metaheuristic is acceptable.

Akhtar et al. (2017) developed a Backtracking Search Algorithm (BSA) in CVRP models considering the smart bin concept to determine the best optimized waste collection routes. Their aims were to minimize the sum of the waste collection route distances. They create significant savings by solving their proposed model in comparison with a conventional condition. Recently, Tirkolaee et al. (2018) developed a MILP model for the multi-trip CARP in order to minimize total cost. They proposed a hybrid algorithm on an improved Max-Min Ant System (IMMAS) to solve well-known test problems and large-sized instances. They could demonstrate the high efficiency of their algorithm.

This paper is a development study for the previous research done by Babaee Tirkolaee et al. (2016) in order to cover its research gaps such as considering multiple trips and drivers and crews working time. The aim is to develop a novel mathematical model for the robust multi-trip PCARP regarding the waste collection problem in order to minimize the total cost (the usage cost of vehicles and traversing cost of the edges) considering the demand uncertainty of the required edges and working time of drivers and crew besides vehicles' capacity constraints in order to study the real world assumptions existed in the problem which has not been considered before. This proposed model constructs a general framework to be applied in each real case. The aim is to determine the optimal number and the optimal routes of the vehicles in each time period.

Since the proposed problem is a complicated version of the classic CARP, it would be an NP-hard problem (Golden and Wong 1981). Hence, a hybrid Simulated Annealing (SA) algorithm is developed using a constructive heuristic and an efficient cooling equation to solve the problem.

2. Model development

The problem is to determine the optimal number of vehicles the optimal tour for each vehicle including possible multiple trips with regard to the objective function which minimizes the usage cost of vehicles and the traversing cost of all traversed edges in all time periods of planning. In this problem, the vehicles are located in the depot, firstly. They begin their tour in order to service the required edges through collecting wastes and then, return to the depot after filling their capacity. The capacity constraint of vehicles and their maximum available times result in determining the number of required vehicles. In fact, drivers and crew's working time denotes the maximum available time considered for each vehicle. Moreover, since the demand parameter is considered to be uncertain, the robust model is presented based on the interval robust optimization approach introduced by Bertsimas and Sim (2004).

2.1. Problem assumptions

- The objective function, in addition to the minimization of traversing cost of the edges, is to minimize the number of required vehicles to meet the total demand.
- Each required edge is serviced only by one vehicle. Also, only one depot is considered in the problem.
- The vehicles are heterogeneous and the graph network is asymmetric. Each vehicle begins the tour from the depot and returns there finally.
- Each type of vehicle has a unique average speed and waste loading time on each edge.
- The traversing time and the waste loading time are different for each vehicle.
- Vehicle usage cost is the cost paying to the driver and related crew for each vehicle.
- Each vehicle has a maximum duration of service denoting the working time of its driver and crew.
- Each vehicle may have multiple trips.
- There is a time horizon for planning. The planning horizon is a week. Each required edge should be serviced every day. The collection planning of each vehicle is determined optimally for each week on a daily basis.
- Each required edge needs to be serviced in each period.
- The demand of required edges fluctuates in a given uncertainty interval.

2.2. Sets, parameters, and variables

In this section, the robust model of the proposed problem is presented. For this purpose, sets, parameters, and variables are defined firstly. Then, the theoretical basis of the robust optimization is described and finally, the robust model is presented.

Sets

V: Set of the nodes in the graph network
K: Set of the vehicles
E: Set of all the edges defined in the network

E_R: Set of all the required edges defined in the network
T: Set of time periods
O: Set of possible multiple trips
S: Optional set of all the edges defined in the network
V [S]: Set of the nodes defined in S

Parameters

dis_{ij}: Traversing distance of the edge (i,j)
Vel_{ijk}: Average speed of kth vehicle for traversing the edge (i,j)
Lw_{ijk}: Waste loading time of kth vehicle on the edge (i,j)
d_{ijt}: Uncertain demand for edge (i,j) in period t
$\bar{d}_{ijt}$: Average demand for edge (i,j) in period t
cv_k: Usage cost of kth vehicle
W_k: Capacity of kth vehicle in each trip
$T_{\max}$: Maximum available time for vehicles in each period
λ: Distance conversion ratio to cost
M: A very large optional number

Decision variables

x^o_{ijkt}: 1 if edge $(i,j) \in E$ is traversed by oth trip of vehicle k in period t, otherwise 0.
y^o_{ijkt}: 1 if edge $(i,j) \in E$ is serviced by oth trip of vehicle k in period t, otherwise 0.
u_{kt}: 1 if vehicle k is used in period t, otherwise 0.

2.3. The robustness of capacity constraints of vehicles

In this section, a robust mathematical model for the classic arc routing problem is formulated based on Bertsimas and Sim approach (2003, 2004). In this model, the uncertain demand d_{ijt} is modeled as a symmetric and bounded random variable that takes value in $[\bar{d}_{ijt} - \hat{d}_{ijt}, \bar{d}_{ijt} + \hat{d}_{ijt}]$, where $\hat{d}_{ijt}$ is the deviation of $\bar{d}_{ijt}$ (Ben-Tal and Nemirovski, 2000). This parameter causes uncertainty in the capacity constraints of vehicles. The corresponding constraints with uncertainty are:

$$\sum_{(i,j)\in E_R} d_{ijt} y^o_{ijkt} \leqslant W_k, \quad \forall k \in K,\ \forall t \in T,\ \forall o \in O \tag{1}$$

So, the parameter Γ_{kto} is defined for each vehicle k in period t such that $\Gamma_{kto} \in [0, |J_{kto}|]$, where J_{kto} is the set of the serviced edges by oth trip of vehicle k in period t and it is defined as a set of uncertain coefficients in the constraint corresponding to the capacity constraint and usage time limitation of vehicle k in each period. In fact, Γ_{kto} ensures the robustness adjustment against the protection level of the solution.

Bertsimas and Sim (2004) showed that it is unlikely that all of the coefficients change simultaneously. Therefore, it is assumed that up to $\lfloor \Gamma_{kto} \rfloor$ of these coefficients are allowed to change, and one coefficient $d_{e_{kto}t}$ (corresponding to the edge e_{kto}) can change up to $(\Gamma_{kto} - \lfloor \Gamma_{kto} \rfloor)\hat{d}_{e_{kto}t}$. Then, the solution still remains feasible. In other words, it is assumed that only a subset of the coefficients is permitted to affect the solution adversely. This assumption assures that if it happens in a real case, the robust optimal solution will be feasible definitely. Moreover, due to the symmetric distribution of variables, even if more than $\lfloor \Gamma_{kto} \rfloor$ changes occur, then, the robust solution will be feasible with a very high probability. Thus, $\lfloor \Gamma_{kto} \rfloor$ is called the conservation level for vehicle k in period t in trip o. On this basis, the robust mathematical model of the proposed nominal linear optimization is as follows:

$$sum_{(i,j)\in E_R} d_{ijt} y^o_{ijkt} + \underset{\{S_{kto}\cup\{e_{kto}\}|S_{kto}\subseteq J_{kto},|S_{kto}|=\lfloor\Gamma_{kto}\rfloor,\ e_{kto}\in J_{kto}\setminus S_{kto}\}}{Max} \times \left\{ \sum_{(i,j)\in S_{kto}} \hat{d}_{ijt}|y^o_{ijkt}| + (\Gamma_{kto} - \lfloor \Gamma_{kto} \rfloor)\hat{d}_{e_{kto}t}|y^o_{e_{kto}kt}| \right\} \leqslant W_k, \forall k \in K,\ \forall t \in T,\ \forall o \in O \tag{2}$$

where

$$\beta_{kto}(Y^*, \Gamma_{kto}) = \underset{\{S_{kto}\cup\{e_{kto}\}\,|\,S_{kto}\subseteq J_{kto},|S_{kto}|=\lfloor\Gamma_{kto}\rfloor,\, e_{kto}\in J_{kto}\setminus S_{kto}\}}{Max} \times \left\{ \sum_{(i,j)\in S_{kto}} \hat{d}_{ijt}|y^{o^*}_{ijkt}| + (\Gamma_{kto} - \lfloor \Gamma_{kto} \rfloor)\hat{d}_{e_{kto}t}|y^{o^*}_{e_{kto}kt}| \right\} \tag{3}$$

Eq. (3) is equal to the optimal value of the problem of Eq. (4) which is linear (For more information, see Bertsimas and Sim (2004)).

$$\beta_{kto}(Y^*, \Gamma_{kto}) = max \sum_{(i,j)\in J_{kt}} \hat{d}_{ijt}|y^{o^*}_{ijkt}|z^o_{ijkt} \sum_{(i,j)\in J_{kto}} z^o_{ijkt} \leqslant \Gamma_{kto}, \quad \forall k \in K,\ \forall t \in T,\ \forall o \in O \quad 0 \leqslant z^o_{ijkt} \leqslant 1, \forall (i,j) \in J_{kto}, \forall k \in K, \quad \forall t \in T,\ \forall o \in O \tag{4}$$

Γ_{kto} can be integer or not. The point is, if $\Gamma_{kto} = 0$, then $\beta_{kto}(Y^*, \Gamma_{kto}) = 0$. Thus, the corresponding robust constraint would be similar to the corresponding constraint of the nominal problem. Likewise, if $\Gamma_{kto} = |J_{kto}|$, Bertsimas and Sim approach will be as well as Soyster's method (Bertsimas and Sim, 2004). Therefore, by changing Γ_{kto} in $[0, |J_{kto}|]$, the robustness adjustment is ensured against the conservatism level of the solution. Anyway, the dual of the problem is as follows:

$$Dual\ \beta_{kto}(Y^*, \Gamma_{kto}) = \min \sum_{(i,j)\in J_{kto}} r^o_{ijkt} + \Gamma_{kto} z_{kto}$$

$$Subject\ To: \quad z_{kto} + r^o_{ijkt} \geqslant \hat{d}_{ijt}|y^{o^*}_{ijkt}|\ \forall k \in K, \forall (i,j) \in J_{kto}, \forall t \in T, \forall o \in O \quad r^o_{ijkt} \geqslant 0, z_{kto} \geqslant 0\ \forall k \in K, \forall (i,j) \in J_{kto}, \forall t \in T,\ \forall o \in O \tag{5}$$

Finally, the linear and robust form of the problem would be:

$$\sum_{(i,j)\in J_{kto}} \bar{d}_{ijt} y^o_{ijkt} + z_{kto}\Gamma_{kto} + \sum_{(i,j)\in J_{kto}} r^o_{ijkt} \leqslant W_k \forall k \in K, \forall t \in T, \forall o \in O \quad z_{kto} + r^o_{ijkt} \geqslant \hat{d}_{ij} E^o_{ijkt} \forall k \in K, \forall (i,j) \in J_{kto},\ \forall t \in T, \quad \forall o \in O, -E^o_{ijkt} \leqslant y^o_{ijkt} \leqslant E^o_{ijkt} \forall k \in K, \forall (i,j) \in J_{kto},\ \forall t \in T, \quad \forall o \in O, r^o_{ijkt} \geqslant 0, z_{kto} \geqslant 0\ \forall k \in K, \forall (i,j) \in J_{kt},\ \forall t \in T,\ \forall o \in O. \tag{6}$$

As it is assumed that all of the required edges are uncertain, so, the set J_{kto} includes a subset of the required edges (E_R) which are being serviced by vehicle k in period t. It is worthy to say that the robustness variables of (z_{kto}, r^o_{ijkt}, E^o_{ijkt}) are used in order to adjust the robustness of the solutions and applying the protection levels to the model.

2.4. Proposed robust model

In this section, the proposed robust mathematical model based on the developed model in (Babaee Tirkolaee et al., 2016) is presented:

$$MinimizeCost_{total} = \lambda\Big(\sum_{(i,j)\in E}\sum_{k\in K}\sum_{o\in O}\sum_{t\in T} dis_{ij} x^o_{ijkt}\Big) + \sum_{k\in K}\sum_{t\in T} cv_k u_{kt}$$

$$Subject\ to: \tag{7}$$

$$\sum_{(i,j)\in E} x^o_{ijkt} = \sum_{(j,i)\in E} x^o_{jikt} \quad \forall i \in V, \forall k \in K, \forall t \in T, \forall o \in O, \tag{8}$$

$$\sum_{k\in K}(y^o_{ijkt}+y^o_{jikt})=1 \quad \forall(i,j) \text{ or}(j,i)\in E_R, \quad \forall t\in T, \forall o\in O, \tag{9}$$

$$\sum_{(i,j)\in E_R}\bar{d}_{ijt}y^o_{ijkt}+z_{kto}\Gamma_{kto}+\sum_{(i,j)\in E_R}r^o_{ijkt}\leqslant W_k \quad \forall k\in K, \forall t\in T, \forall o\in O, \tag{10}$$

$$z_{kto}+r^o_{ijkt}\geqslant \hat{d}_{ijt}E^o_{ijkt} \quad \forall k\in K, \forall(i,j)\in E_R, \forall t\in T, \forall o\in O, \tag{11}$$

$$-E^o_{ijkt}\leqslant y^o_{ijkt}\leqslant E^o_{ijkt} \quad \forall k\in K, \forall(i,j)\in E_R, \forall t\in T, \forall o\in O, \tag{12}$$

$$y^o_{ijkt}\leqslant x^o_{ijkt} \quad \forall(i,j)\in E, \forall k\in K, \forall t\in T, \forall o\in O, \tag{13}$$

$$\sum_{(i,j)\in E}\sum_{o\in O}x^o_{ijkt}\leqslant M u_{kt} \quad \forall k\in K, \forall t\in T, \tag{14}$$

$$\sum_{(i,j)\in E}\sum_{o\in O}\frac{dis_{ij}}{Vel_{ijk}}x^o_{ijkt}+\sum_{(i,j)\in E_R}\sum_{o\in O}Lw_{ijk}y^o_{ijkt}\leqslant T_{\max} \quad \forall k\in K, \forall t\in T, \tag{15}$$

$$\sum_{i,j\in S}x^o_{ijkt}\leqslant 1+nh^s_{kto} \quad , \forall S\subseteq V\setminus\{1\}; S\neq\varnothing; \forall k\in K, \forall t\in T, \forall o\in O, \tag{16}$$

$$\sum_{i\in S}\sum_{j\neq S}x^o_{ijkt}\geqslant 1-f^s_{kto} \quad \forall S\subseteq V\setminus\{1\}; S\neq\varnothing; \forall k\in K, \forall t\in T, \forall o\in O, \tag{17}$$

$$h^s_{kto}+f^s_{kto}\leqslant 1 \quad \forall S\subseteq V\setminus\{1\}; S\neq\varnothing; \forall k\in K, \forall t\in T, \forall o\in O, \tag{18}$$

$$h^s_{kto}\in\{0,1\}, f^s_{kto}\in\{0,1\} \quad \forall S\subseteq V\setminus\{1\}; S\neq\varnothing; \forall k\in K, \forall t\in T, \forall o\in O, \tag{19}$$

$$x^o_{ijkt}\in\{0,1\}, y^o_{ijkt}\in\{0,1\}, u_{kt}\in\{0,1\} \quad \forall(i,j)\in E, \forall k\in K, \forall t\in T, \forall o\in O, \tag{20}$$

$$E^o_{ijkt}, z_{kto}, r^o_{ijkt}\geqslant 0 \quad \forall(i,j)\in E, \forall k\in K, \forall t\in T, \forall o\in O. \tag{21}$$

The objective function includes minimizing the total traversing and usage cost of the vehicles in all periods. Constraints (8) are the flow balance equations for each vehicle in each period. These constraints control the input and output flow of the edges (i.e. if a vehicle enters a node, it must leave it necessarily). Constraints (9) ensure that any required edges must be serviced in each period just once (but it can be traversed more than one time). Constraints (10) to (12) are the corresponding constraints to the robustness. Constraints (13) express that a required edge will be serviced by the vehicle which has traversed the edge (in other words, the vehicle may traverse the edge without servicing). Constraints (14) ensure that kth vehicle will be used, whenever its usage cost is paid. Constraints (15) explain that total traversing time of each vehicle should not exceed the maximum available time considered. Constraints (16) to (18) are the sub-tour elimination constraints (Eislet and Laporte, 1995) where h^s_{kto} and f^s_{kto} are relevant non-decision binary variables used in sub-tour elimination constraints. Constraints set (19)–(21) specify the types of variables.

2.5. Determining a probability bounds for constraint violation

Given a value of uncertainty budget Γ_{kto}, the model (7)–(21) generates solutions that ensures 100% of feasibility if at most Γ_{kto} random parameter of demand change. However, if more than Γ_{kto} random parameters change, the generated solution will be feasible with high probability (Bertsimas and Sim, 2004). Bertsimas and Sim (2004) proposed different approaches to determine upper bounds related to the violation probability of the problem constraints. One of those approaches requires little computational effort.

In the proposed robust model, if the probability of violating the capacity constraint for each vehicle in its trip and defined period is α_{kto}, then the corresponding constraint to the capacity for each vehicle must satisfy the following equation:

$$Pr\left(\sum_{(i,j)\in E_R}d_{ijt}y^o_{ijkt}>W_k\right)\leqslant\alpha_{kto} \text{ or } Pr\left(\sum_{(i,j)\in E_R}d_{ijt}y^o_{ijkt}\leqslant W_k\right)\geqslant(1-\alpha_{kto}) \tag{22}$$

In continuation, a proposition is presented which is used for determining a bound for the above probability (Bertsimas and Sim, 2004).

The probability that the constraint is violated holds true in (For more information, see (Bertsimas and Sim, 2004)):

$$Pr\left(\sum_{(i,j)\in E_R}d_{ijt}y^o_{ijkt}>W_k\right)\leqslant B(n,\Gamma_{kto}),\ \alpha_{kto}=B(n,\Gamma_{kto}) \tag{23}$$

where

$$\begin{aligned}B(n,\Gamma_{kto})&=\frac{1}{2^n}\left\{(1-\mu)\sum_{l=\nu}^{n}\binom{n}{l}+\mu\sum_{l=\nu+1}^{n}\binom{n}{l}\right\}\\&=\frac{1}{2^n}\left\{(1-\mu)\binom{n}{\nu}+\sum_{l=\nu+1}^{n}\binom{n}{l}\right\}\end{aligned} \tag{24}$$

where $n=|J_{kto}|$, $\nu=\frac{\Gamma_{kto}+n}{2}$ and $\mu=\nu-\lfloor\nu\rfloor$. Therefore, for the proposed problem, firstly the number of serviced edges in each route is specified (n) to control the violation probability of the capacity constraints. Then, Γ_{kto} is determined based on the desired service level ($1-\alpha_{kto}$) and the proven equation. Thus, Γ_{kto} should be greater than the least value which satisfies Eq. (24).

3. Solution method

Due to the high degree of the problem complexity and also, considering uncertainty in the constraints of the proposed model, solving the problem by the exact method is only possible in small size problems. Therefore, in this section, a hybrid algorithm based on a heuristic algorithm and Simulated Annealing (SA) algorithm is developed to solve the problem approximately. The heuristic algorithm is implemented to generate some initial solutions and then, the SA algorithm is applied to provide the best feasible solutions. The CPLEX solver of GAMS software is used to solve the problem exactly and to evaluate the efficiency of the proposed algorithm. In the next sections, the mechanism of the proposed algorithm will be discussed.

3.1. Solution representation

The solution representation method has direct effects on the quality of results in metaheuristic algorithms. In this paper, a matrix is used to represent the traversing sequence of the required edges besides the relevant vehicle number, trip number and period number. In other words, the applied chromosome here is a sequence of the required edges related to the corresponding vehicle, vehicle trip and period. Each chromosome consists of seven cells in which there are three separators of zero between vehicle number and period number, between period number and trip number, and between trip number and constructed routes, respectively. The first cell of the chromosome represents the vehicle number, the third represents the period number, the fifth

represents the trip number of the vehicle, and the last cell contains the constructed route for that specific vehicle, trip and period. Even numbered cells represent the separators. See Table 1.

First, the vehicle begins its first trip from the depot and returns there after serving the specified required edges in the string (after filling out the capacity). Then, if possible (due to the operational limitations), the vehicle begins its next trip from the depot and services the next remained required edges in the string. Otherwise, the next vehicle begins its first trip and enters the operational area in order to serve the rest of the required edges. This will continue until all the required edges are being serviced. Note that the traversing cost between two required edges is the shortest path between them which has been determined by Dijkstra algorithm before. In other words, the shortest path between two required edges is equal to the shortest path between the end node constituting the first edge and the first node constituting the second edge. For example, the shortest path between two edges of (2, 3) and (5, 7) is equal to the shortest path between nodes 3 and 5.

Fig. 1 shows an instance problem which includes 7 nodes, 13 edges, 6 required edges and two possible types of vehicle with two different time periods (even numbered days and odd numbered days of the week). The string shown in Table 1 includes the solution with one used vehicle and four chromosomes. The first chromosome is assigned to the first trip of the first vehicle in its first planning period, the second is assigned to the second trip of the first vehicle for its first period planning, the third is assigned to the first trip of the first vehicle in its first planning period, and the last one is assigned to the second trip of the first vehicle in its first planning period.

3.2. The heuristic algorithm

To generate the initial solutions, a constructive heuristic algorithm is applied with detailed execution steps in the following:

Step (1) – If there is any time period to be planned, select a period among the existing periods randomly and go to Step (2). Otherwise, go to Step (6).
Step (2) – Randomly select a vehicle among the existing vehicles and start its first trip from the depot and go to Step (3).
Step (3) – Among the required edges which their starting nodes are the end node of the last added edge, consider k- required edges for which the capacity and time limitations of the vehicle allows to be added to the trip and select one edge with the least distance from the depot and go to Step (5). Otherwise, if there is no required edge observing one of these conditions, select an edge with the least distance that can be added to the last selected edge, and go to Step (4).
Step (4) – If time limitation allows the vehicle to have another trip, start the next trip from the depot and go to Step (3), otherwise, go to Step (5).
Step (5) – If there is a required edge to be served, go to Step (3). Otherwise, select the shortest path to the depot and go to Step (1).
Step (6) – Report the solution obtained.

The number of generated solutions is aimed to be 200. When the heuristic algorithm ends, the best solution is used for SA algorithm.

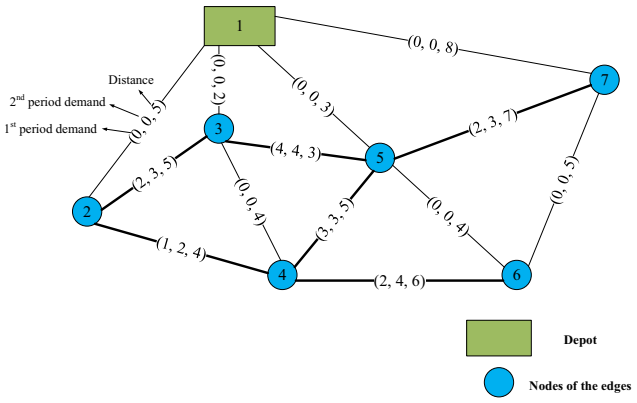


Fig. 1. A schematic instance problem.

3.3. Simulated annealing algorithm

In this research, simulated annealing algorithm is applied to improve the quality of the solutions (Kirkpatrick and Vecchi 1983). As a consequence, all the initial solutions are improved individually by this algorithm. The general framework of SA is such that it starts with an initial solution. The initial parameters of the algorithm include the number of iterations in each temperature (M), initial temperature (t_0), cooling rate (β), final temperature (t_{end}) and Boltzmann constant (K) which are initialized before the search starts. Then, a neighborhood is created for the initial solution and it will be replaced with the initial solution if its objective value is better than the initial solution. Otherwise, the difference between two objective values is calculated (Δ) and a random number is generated from the interval [0, 1]. In continuation, the randomly generated number is compared with the probability equation of the algorithm ($\exp(-\Delta/KT)$) and the worse solution is accepted if the random number is less than the value of probability equation. In each temperature, a number of iterations are run and then, the temperature is reduced. The temperature reduction (cooling) equation is described in Eq. (25) (Glover and Kochenberger, 2006). In addition, stopping criterion is to reach the final temperature.

$$t_{i+1} = \lim_{\beta \to 0} \frac{t_i}{1 + \beta t_i} \tag{25}$$

where β is a small optional value. The number of iterations for each temperature is obtained from Eq. (26).

$$M(t_i) = \frac{t_i}{1 + \eta t_i} \tag{26}$$

In the present paper, due to the randomized selection of three problems as instances, trial-and-error method is used to define the values of algorithm parameters as shown in Table 2.

The local searches have been used in this algorithm are as below:

1. Replace the planned period given for each trip of the vehicle by another vehicle.

Table 1
Solution representation matrix.

Description	Vehicle number	S	Period number	S	Trip number	S	Constructed routes
First trip of the first vehicle in period 1	1	0	1	0	1	0	1–3–5–4–6–7–1
Second trip of the first vehicle in period 1	1	0	1	0	2	0	1–3–2–4–5–7–1
First trip of the first vehicle in period 2	1	0	2	0	1	0	1–5–3–2–4–5–1
Second trip of the first vehicle in period 2	1	0	2	0	2	0	1–3–4–6–5–7–1

Table 2
Values of algorithm parameters.

Parameters	Values
α	0.98
β	0.0001
t_0	200
t_{end}	1
η	0.1

2. Two trips of two different vehicles are selected randomly. If there is a mutual edge/edges between trips, one of them is selected randomly and so, both tours will be divided into the two parts accordingly. The first part of the first trip is combined with the second part of the second trip and the two other components of the trips are combined with each other. As a result, two new trips are composed.
3. Two trips are selected randomly. If there are two mutual edges between them, the sequences between the two mutual edges are changed together.
4. An edge of a trip is selected randomly and its direction is reversed.
5. A part of a trip is selected randomly and their directions are changed.

Finally, all the solutions are checked to be feasible. All the infeasible solutions will be removed.

3.4. Calculating fitness function of the robust generated solutions

To determine the fitness function value of each generated solution by metaheuristic method, firstly the objective value (which is the total cost of traversed distance by vehicle) is calculated regardless of the solution robustness. Then, the following steps are done:

Step (1). The numbers of required edges in each constructed tour are considered. Then, on the basis of Eqs. (23) and (24), the service level is defined and Γ_{kto} is determined for the vehicles.
Step (2). Γ_{kto} of the edges with the highest uncertainty value of demand are selected and their demands are increased to the maximum possible value, according to their demand uncertainty interval.
Step (3). If Γ_{kto} is chosen as a non-integer, the demand of the last selected edge is increased by $(\Gamma_{kto} - \lfloor\Gamma_{kto}\rfloor)\widehat{d}_{ijt}$, where $\widehat{d}_{ijt}$ is defined by multiplying the demand parameter by the deviation level δ (i.e. $\widehat{d}_{ijt} = \delta d_{ijt}$).
Step (4). With regard to the considered changes in the demand of the edges, if the demand of each edge exceeds the vehicle capacity and its usage time limit, a penalty proportional to the violation value is added to the objective function.

3.5. Examination of the solutions obtained for the same task

Since SA algorithm is a non-deterministic procedure (Hasançebi et al., 2010), thus it is a necessary and applicable issue to examine a set of solutions obtained for the same task. In fact, SA can report all these feasible solutions with respect to its operational limitations. In this case, the objective function value is almost the same for all such solutions, but the details of solutions could differ such as the vehicles used, the priority of serving required arcs and etc. In other words, SA can provide such solutions due to its nature and the values of model parameters in different problems. In real cases, the most practical and applicable solution can be chosen and implemented.

4. Model results

In this section, the method of generating different random instances for evaluating the performance of the proposed algorithm is described. Also, the efficiency of the algorithm is studied in comparison with CPLEX solver as the exact method.

4.1. Model verification

Model verification is studied by generating and solving some instance problems in different sized. To generate the random instances in different sizes, n nodes are generated randomly in a two-dimensional coordinate space and node 1 is considered as the depot. Then, a $n \times n$ matrix is considered to define the available edges in the instances. The value of each cell is 1 by the probability of 0.7 or 0 by the probability of 0.3. In this matrix, number 1 indicates that the edge exists in the instance. Then, up to 80% of the edges are selected randomly as to be required edges and their demands are generated randomly using uniform distribution U [5,10]. Two types of vehicles are considered with different capacity and usage cost. $cv_{k=1}$, $cv_{k=2}$, $Vel_{k=1}$, $Vel_{k=2}$, $W_{k=1}$, $W_{k=2}$, and $T_{\max}$ are considered to be 500$, 800$, 20 km/h, 30 km/h, 500 kg, 1000 kg and 4 h, respectively. Finally, c_{ij} and Γ_{kto} are generated randomly using uniform distribution U [0.1, 2] and U [1.1, 10] in each instance problems. Input data about sets are given in Table 3. 10 instance problems are generated to validate the model and test the algorithm in comparison with CPLEX. Also, two additional instance problems are generated in large size to estimate the number of edges possible to be solved using the proposed algorithm.

In Table 3, n, TE, and RE are the numbers of nodes, total edges, and total required edges in the network graphs, respectively. Also, AV1 and AV2 are the number of available vehicles type 1 and type 2, respectively. The results obtained by solving the robust model by the proposed algorithm and CPLEX solver, are presented in Tables 4–7 and for two different deviation level of 0.05, 0.075, 0.1 and 0.2.

Table 3
Random generated instances.

Problem	n	TE	RE	AV1	AV2
P_1	6	10	8	1	1
P_2	8	16	13	2	1
P_3	9	22	16	2	2
P_4	11	34	25	3	2
P_5	12	40	29	4	3
P_6	14	52	35	4	4
P_7	15	64	46	5	4
P_8	16	75	55	6	5
P_9	17	85	61	7	6
P_{10}	18	100	72	10	8
P_{11}	25	150	94	12	10
P_{12}	30	280	180	16	14

Table 4
Comparing results of SA and CPLEX for robust problem with $\widehat{d} = 0.05d$.

Problem	CPLEX		SA		
	Objective function	Run time (sec)	Objective function	Run time (sec)	GAP (%)
P_1	740.93	86.94	757.59	11.61	2.25
P_2	919.5	173.74	928.03	23.99	0.93
P_3	1406.74	600.54	1441.65	30.27	2.48
P_4	2161.53	856.23	2214.27	31.53	2.44
P_5	2739.57	1379.17	2798.96	48.35	2.17
P_6	2998.99	2015.8	3061.44	50.8	2.08
P_7	3919.52	3209.14	3978.27	69.54	1.5
P_8	4826.8	3600	4938.75	96.39	2.32
P_9	–*	3600	5884.43	117.93	–
P_{10}	–	3600	6888.68	191.58	–

* No solution found.

Table 5
Comparing results of SA and CPLEX for robust problem with $\widehat{d} = 0.075d$.

Problem	CPLEX		SA		
	Objective function	Run time (sec)	Objective function	Run time (sec)	GAP (%)
P_1	770.3	109.6	785.65	14.81	1.99
P_2	1277.1	262.94	1308.18	19.51	2.43
P_3	2045.62	531.23	2073.01	47.67	1.34
P_4	2481.21	1321.72	2568.53	59.2	3.52
P_5	2706.62	1679.23	2792.97	65.47	3.19
P_6	3562.13	2727.69	3622.18	71.85	1.69
P_7	4022.34	3600	4057.01	92.66	0.86
P_8	–*	3600	5256.41	81.43	–
P_9	–	3600	6369.36	151.82	–
P_{10}	–	3600	7970.87	201.33	–

* No solution found.

Table 6
Comparing results of SA and CPLEX for robust problem with $\widehat{d} = 0.1d$.

Problem	CPLEX		SA		
	Objective function	Run time (sec)	Objective function	Run time (sec)	GAP (%)
P_1	882.06	117	916.09	15.56	1.2
P_2	1575.88	687.52	1595.5	33.55	1.26
P_3	2431.12	851.39	2498.96	68.22	0.95
P_4	3086.07	1592.81	3143.47	68.6	1.86
P_5	3727.73	2388.79	3808.25	71.51	2.16
P_6	4425.04	3359.2	4518.03	95.37	1.75
P_7	5075.26	3600	5174.13	123.65	1
P_8	–*	3600	6674.97	152.61	–
P_9	–	3600	8493.2	183.63	–
P_{10}	–	3600	10163.17	295.68	–

* No solution found.

Table 7
Comparing results of SA and CPLEX for robust problem with $\widehat{d} = 0.2d$.

Problem	CPLEX		SA		
	Objective function	Run time (sec)	Objective function	Run time (sec)	GAP (%)
P_1	1050.4	119.79	1056.52	19.85	0.58
P_2	1732.46	792.1	1777.26	29.8	2.59
P_3	2846.73	1134.17	2919.27	43.83	2.55
P_4	3768.49	1908.56	3817.02	67.12	1.29
P_5	3977.51	3164.53	4045.63	111.11	1.71
P_6	4880.29	3600	4932.02	119.6	1.06
P_7	–*	3600	5939.88	137.97	–
P_8	–	3600	7169.74	169.37	–
P_9	–	3600	12207.51	230	–
P_{10}	–	3600	16490.78	328.65	–

* No solution found.

A run time limitation of 3600 s is considered as the stopping criteria and the best obtained solution during this time is reported.

The GAP between the results of the proposed algorithm and CPLEX is calculated by Eq. (27).

$$GAP = (\text{SA } objective - \text{CPLEX } objective)/\text{CPLEX } objective \quad (27)$$

As it is obvious in the Tables 4–7, CPLEX is unable to find a solution for some problems in 3600 s run time. Results have shown that the proposed algorithm has appropriate efficiency in a reasonable run time. SA could solve the problems at a lower run time of 200 s. Also, average gaps calculated for the random instances with $\widehat{d} = 0.05d$, $\widehat{d} = 0.075d$, $\widehat{d} = 0.1d$ and $\widehat{d} = 0.2d$ are 1.63% and 1.45%, 2.15% and 2.02%, respectively.

A comparison of the run times of the proposed algorithm with the exact method is shown in Fig. 2 for $\widehat{d} = 0.05d$ as an example.

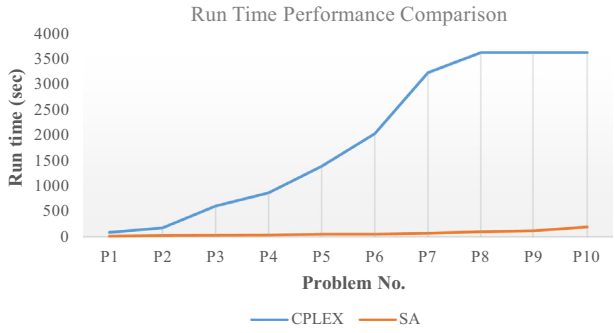


Fig. 2. Run Time performance comparison of the solution methods.

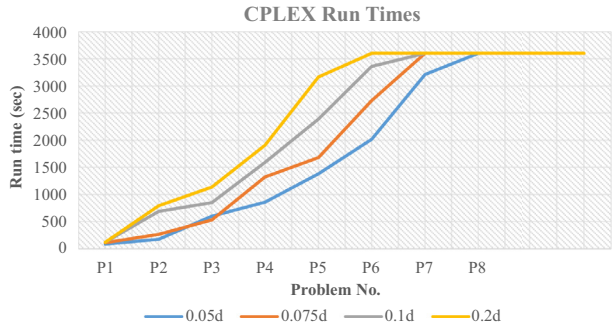


Fig. 3. CPLEX run times under different deviation levels.

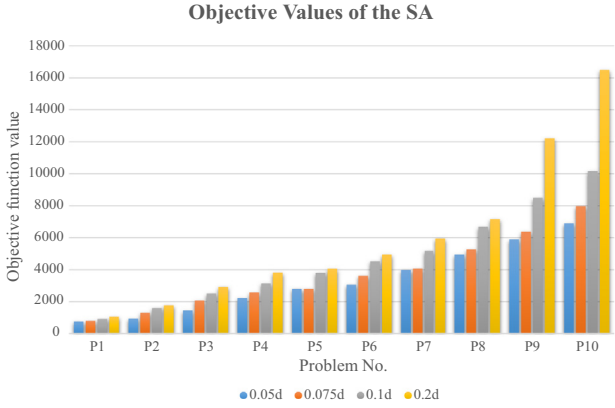


Fig. 4. Objective values for different deviation levels.

Table 8
Capability evaluation of the proposed SA in solving large sized problems with $\widehat{d} = 0.05d$.

Problem	SA	
	Objective function	Run time (sec)
P_{11}	12127.52	851.28
P_{12}	–*	3600

* No solution found.

For all different deviation levels, SA has a great difference with CPLEX in terms of run time.

As it is clear in Fig. 2, the size of the problem does not affect the run time of the proposed algorithm significantly. However, by increasing the size of the problem, the run time of the CPLEX increases with a high slope. But, the run time of the proposed algorithm increases approximately by a linear trend.

Fig. 3 illustrates the effects of the different deviation levels on the CPLEX run time. Also, Fig. 4 depicts the evaluation of the different deviation levels in the problems by their objective values obtained by SA. By increasing the deviation level from 0.05 to 0.2, the objective value of random instances increases with a growing rate such that it has its maximum value at deviation level of 0.2 in each problem. Moreover, the maximum increase of objective value occurs in Problems 9 and 10 which are more significant. It is noticeable that the deviation level plays an important role in the robust problem. In other words, management needs to pay more cost to have a better control on the problem in uncertain conditions.

Finally the capability of the proposed algorithm is tested by solving Problems 11 and 12 $\widehat{d} = 0.05d$ (Table 8). As it is obvious, SA is able to find the best possible solution with the objective value of 12127.52 units and the run time of 851.28 s. For the Problem 12, it is unable to find a solution within 3600 run time limitation. It shows that some additional modification or operators may be used in order to improve the efficiency of the algorithm in order to solve the very large sized problems.

5. Conclusion

The optimal scheduling and routing of vehicles in periodic capacitated arc routing problems is one of the major decisions in organizations such as municipalities in the context of the urban waste collection. The aim is to optimize total cost in planning and scheduling of vehicles in terms of reducing the number of required vehicles and determining the optimal routes. Difficulty in estimating the parameters of the waste collection problems causes uncertainty. In this paper, a robust mixed-integer linear programming model is developed for periodic capacitated arc routing problem considering drivers and crew's working time in order to study the effects of the uncertain demand. To solve large size problems, a hybrid algorithm is applied based on a constructive heuristic algorithm and simulated annealing algorithm with an efficient cooling equation. The results of solving different problems in comparison with CPLEX shows that the proposed metaheuristic has a great efficiency. On the other hand, it is concluded that the problem is sensitive to the deviation level changes of uncertain demand and whatever the size of the problem is larger, the objective values' changes are greater and have a direct relationship with the magnitude of deviation levels. As a general result, we could provide some applicable results according to the model of the integrated problem by considering working times, planning horizon and uncertainty of demand in order to cover the gaps in the literature and create a managerial tool to be helpful in decision making processes.

For future studies, we propose to consider another objective such as environmental objective and service time objective. Also, we can add some other real assumptions like considering the possibility of multiple trips for vehicles.

A: Definition of the robust model notations

The mathematical notations related to the robust model proposed by Bertsimas and Sim (2004) are defined below:

Y^*	The optimal vector of variable y^o_{ijkt}
δ	Deviation level of the demand parameter
$y^{o^*}_{ijkt}$	The optimal value for y^o_{ijkt}
z^o_{ijkt}, r^o_{ijkt}	These variables are used to quantify the protection level against possible variations of the uncertain parameter d_{ijt}
E^o_{ijkt}	It is equal to $\lvert y^{o^*}_{ijkt}\rvert$
Γ_{kto}	It is called the uncertainty budget, defines the maximum number of customers belonging to oth trip of the vehicle k in period t whose demand can deviate from their nominal value
J_{kto}	is the set of the serviced edges by oth trip of vehicle k
S_{kto}	It is the subset of J_{kto} that is equal to $\lfloor \Gamma_{kto} \rfloor$
e_{kto}	It is the subset of J_{kto} such that $e_{kto} \in J_{kto} \setminus s_{kto}$
$\beta_{kto}(Y^*, \Gamma_{kto})$	Protection function

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